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HT Board Booster

Class X Science

Chemistry

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 3 - 4

Session 2026-27

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Class X Chemistry
CBSE SQP 2019-2025
Ch- Metals and Non-Metals

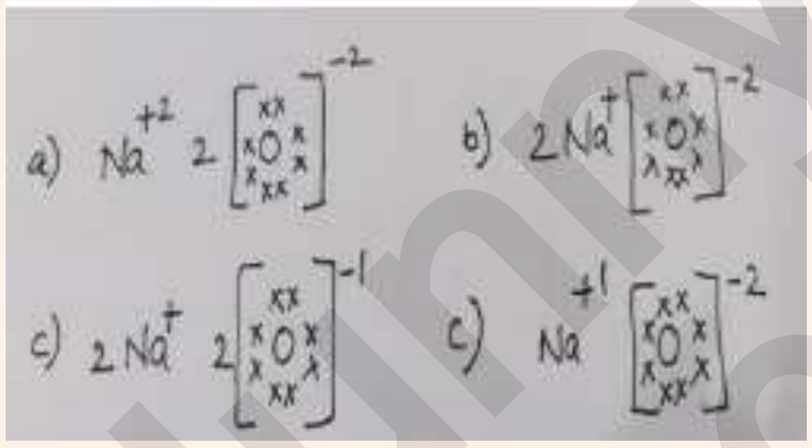
SECTION –A

Q.1. **Assertion:** After white washing the walls, a shiny white finish on walls is obtained after two to three days.
Reason: Calcium Oxide reacts with Carbon dioxide to form Calcium Hydrogen Carbonate which gives shiny white finish. **(CBSE SQP 2020-21)**

Q.2. On placing a copper coin in a test tube containing green ferrous sulphate solution, it will be observed that the ferrous sulphate solution **(CBSE SQP 2022-23)**
(a) turns blue, and a grey substance is deposited on the copper coin.
(b) turns colourless and a grey substance is deposited on the copper coin.
(c) turns colourless and a reddish-brown substance is deposited on the copper coin.
(d) remains green with no change in the copper coin.

Q.3. On adding dilute sulphuric acid to a test tube containing a metal 'X', a colourless gas is produced when a burning match stick is brought near it. Which of the following correctly represents metal 'X'? **(CBSE SQP 2023-24)**
(a) Sodium (b) Zinc (c) Copper (d) Silver

Q.4. Which one of the following correctly represents Sodium oxide? **(CBSE SQP 2023-24)**



Q.5. An element with atomic number _____ will form a basic oxide. **(CBSE SQP 2023-24)**
(a) 7 (2,5) (b) 17 (2,8,7) (c) 14 (2,8,4) (d) 11 (2,8,1)

Q.6. An element 'M' has 50% of the electrons filled in the 3rd shell as in the 2nd shell. The atomic number of 'M' is: **(CBSE SQP 2023-24)**
(a) 10 (b) 12 (c) 14 (d) 18

Q.7. Match column I with column II and select the correct option using the given codes.

Column I	**Column II** **(CBSE SQP 2024-25)**
a. A metal that forms amphoteric oxides	(i) Ga
b. A metal which melts when kept on our palm	(ii) Au
c. A metal that reacts with nitric acid	(iii) Al
d. A metal which cannot displace hydrogen from acids	(iv) Mn

(a) a- (ii), b- (i), c- (iii), d- (iv) (b) a- (iii), b- (i), c- (iv), d- (ii)
(c) a- (iv), b- (ii), c- (iii), d- (i) (d) a- (iii), b- (ii), c- (i), d- (iv)

Q.8. The electronic configuration of three elements X, Y and Z are X- 2, 8, 7; Y-2, 8, 2; and Z - 2,8.
(a) Y and Z are metals (b) Y and X are non-metals **(CBSE SQP 2024-25)**
(c) X is a non-metal and Y is a metal (d) Y is a non-metal and Z is a metal

Q.9. **Assertions (A):** On adding dil. HCI to a test tube containing a substance 'X', a colourless gas is produced which gives a pop sound when a burning match stick is brought near it.
Reason (R): In this reaction metal 'X' is displaced by Hydrogen. **(CBSE SQP 2024-25)**

Q.10. Four statements about the reactions of oxides with dilute hydrochloric acid and aqueous sodium hydroxide are listed. **Which statements are correct?** **(CBSE SQP 2025-26)**

I. Aluminium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
II. Calcium oxide reacts with dilute hydrochloric acid and aqueous sodium hydroxide.
III. Zinc oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
IV. Sulphur dioxide does not react with either dilute hydrochloric acid or aqueous sodium hydroxide.

(a) I and II (b) I and III (c) II and IV (d) III and IV

Q.11. An iron nail is added to each of the two test tubes 'P' and 'Q' containing aqueous copper (II) sulphate, and aqueous silver nitrate respectively. Which of the following observation is correct? **(CBSE SQP 2025-26)**

(a) In test tube 'P' iron nail is coated with a blue coating and in test tube 'Q' there is no reaction.
(b) Iron nail is coated with a brown coating in test tube 'P' and silver coating in test tube 'Q'.
(c) There is no reaction in either of the test tubes 'P' or 'Q'.
(d) There is no reaction in test tube 'P' but a silver coating on iron nail is seen in test tube 'Q'.

Q.12. When excess of carbon dioxide is passed through lime water, the milkiness disappears because: **(CBSE SQP 2025-26)**

(a) water soluble calcium carbonate converts to water soluble calcium bicarbonate.
(b) insoluble calcium carbonate converts to water soluble calcium bicarbonate.
(c) water soluble calcium carbonate converts to insoluble calcium bicarbonate.
(d) insoluble calcium carbonate converts to insoluble calcium bicarbonate.

Section-B

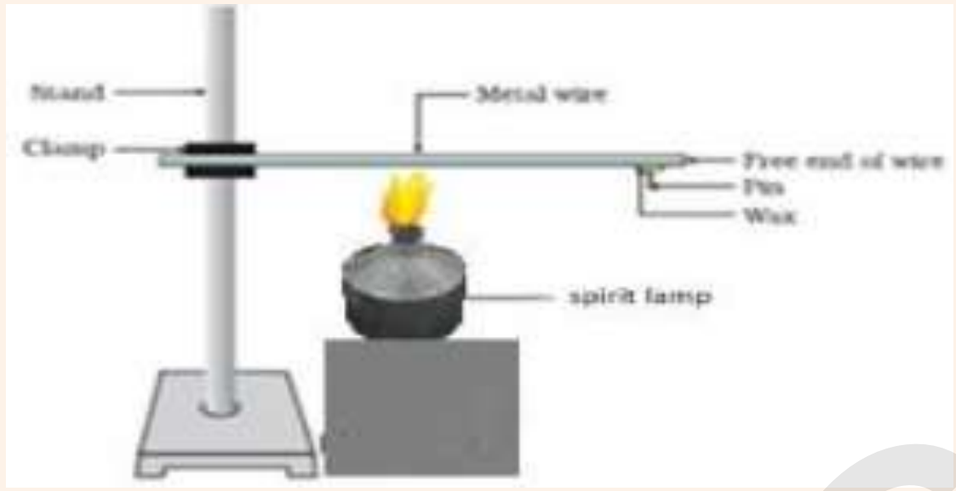
Q.13. The following observations were made by a student on treating four metals P, Q, R and S with the given salt solutions: **(CBSE SQP 2020-21)**

Sample	$MgSO_4(aq)$	$Zn(NO_3)_2(aq)$	$CaSO_4(aq)$	$Na_2SO_4(aq)$
P	No reaction	Reaction occurs	Reaction occurs	No reaction
Q	Reaction occurs	Reaction occurs	Reaction occurs	Reaction occurs
R	No Reaction	Reaction occurs	No Reaction	No Reaction
S	No Reaction	No Reaction	No Reaction	No Reaction

Based on the above observations:
(a) Arrange the given samples in the increasing order of reactivity
(b) Write the chemical formulae of products formed when Q react with$CuSO_4$ solution.

Q.14. Chlorine gas was prepared using electrolysis of brine solution. Write the chemical equation to represent the change. Identify the other products formed in the process and give one application of each. **(CBSE SQP 2023-24)**

Q.15. The following activity is set-up in the science lab by the teacher.
He clamped an aluminium wire on a stand and fixed a pin to the free end of the wire using wax. Then he heated the wire with a burner from the end where the wire is clamped. Students observed the pin fall off.



(a) If the teacher replaces aluminium wire by silver wire, will the students' observation change? Justify your answer.

(b) Will the aluminium wire melt? Give reason for your answer. **(CBSE SQP 2025-26)**

Section-C

Q.16. (a) Explain the formation of Calcium Chloride with the help of electron dot structure. (At numbers: Ca = 20; Cl = 17) **(CBSE SQP 2020-21)**

(b) Why do ionic compounds not conduct electricity in solid state but conduct electricity in molten and aqueous state?

Q.17. The given reaction shows one of the processes to extract the metals like Iron and Manganese.

$$MnO_2\ (s) + Al(s) \rightarrow Mn\ (l) + Al_2O_3\ (s) + Heat$$ **(CBSE SQP 2023-24)**

(a) Give reason why the above reaction is known as a *thermite reaction*.

(b) Identify the substance oxidised and reduced in the above reaction.

(c) Give a reason why Aluminium is preferably used in thermite reactions.

Q.18. An element 'M' with electronic configuration 2 8 3 combines separately with Cl^-, $SO4^{-2}$ anions. Write the chemical formulae of the compounds formed. Predict with the suitable reason the nature of the bond formed by element 'M' in general. How will the electrical conductivity of the compounds formed vary with respect to 'M'? **(CBSE SQP 2023-24)**

OR

A reddish-brown metal 'X', when heated in air, gives a black compound 'Y', which when heated in presence of H_2 gas gives 'X' back. 'X' is refined by the process of electrolysis; this refined form of 'X' is used in electrical wiring.
Identify 'X' and 'Y'. Draw a well – labeled diagram to represent the process of refining 'X'.

Q.19. (a) Anirudh took two metal oxides; aluminium oxide and magnesium oxide as shown in the pictures given below. But he forgot to label them. How will you guide/ help Anirudh to identify the oxides and label them? **(CBSE SQP 2024-25)**





(b) In an activity Aishu was given two substances; Copper Sulphide (Cu_2S) and Copper Oxide (Cu_2O) to obtain copper from these compounds. She was able to extract Copper successfully. Illustrate with the help of chemical equations how Aishu might have completed the activity.

Q.20. **Attempt either option A or B.**

A. An element 'X' is stored in kerosene, and cannot be extracted from its ore using a reducing agent. 'X' forms an ionic compound on reaction with chlorine.

(i) Can we store 'X' in water? Give reason to support your answer.

(ii) Identify element 'X'. Name the process used and write the equation for extraction of 'X' from its ore.

OR

B. The domes of many building in Europe are made of copper. These domes now appear greenish in colour.

(i) Why do the domes appear greenish though copper is orange-red in colour?

(ii) In your opinion, should the copper domes be replaced by iron domes to overcome the problem of change of colour of copper domes?

(iii) Domes used to be made from thin sheets of metals. Why did the ancient architects use copper to make domes? **(CBSE SQP 2025-26)**

Section-D

Q.21. Metal X is found in nature as its sulphide XS. It is used in the galvanisation of iron articles. Identify the metal X. How will you convert this sulphide ore into the metal? Explain with equations. **(CBSE SQP 2019-20)**

OR

State the reason for the following:

(i) Aluminium oxide is called an amphoteric oxide.

(ii) An iron strip dipped in a blue copper sulphate solution turns the blue solution pale green.

(iii) Hydrogen gas is not evolved when most metals react with nitric acid.

(iv) Calcium does not occur in free state in nature.

(v) Sodium or potassium metals are kept immersed under kerosene.

Section-E

Q.22. Two students decided to investigate the effect of water and air on iron object under identical experimental conditions. They measured the mass of each object before placing it partially immersed in 10 ml of water. After a few days, the object was removed, dried and their masses were measured. The table shows their results. **(CBSE SQP 2022-23)**

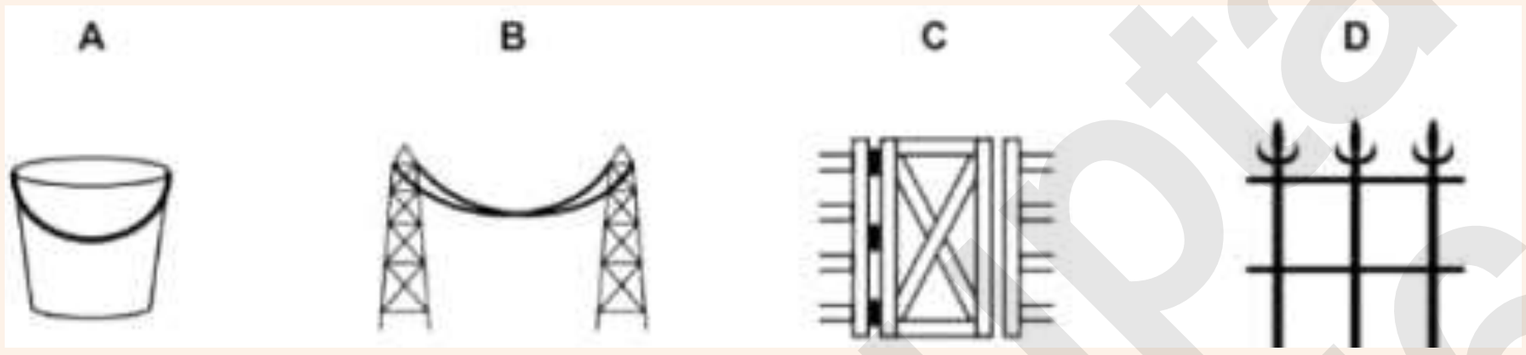
Student	Object	Mass of object before rusting in g.	Mass of the coated object in g.
A	Nail	3.0	3.15
B	Thin plate	6.0	6.33

(a) What might be the reason for the varied observations of the two students?

(b) In another set up the students coated iron nails with zinc metal and noted that, iron nails coated with zinc prevents rusting. They also observed that zinc initially acts as a physical barrier, but an extra advantage of using zinc is that it continues to prevent rusting even if the layer of zinc is damaged. Name this process of rust prevention and give any two other methods to prevent rusting.

OR

(b) In which of the following applications of Iron, rusting will occur most? Support your answer with valid reason.



A - Iron Bucket electroplated with Zinc
B - Electricity cables having iron wires covered with aluminium.
C - Iron hinges on a gate
D - Painted iron fence

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Class X Chemistry

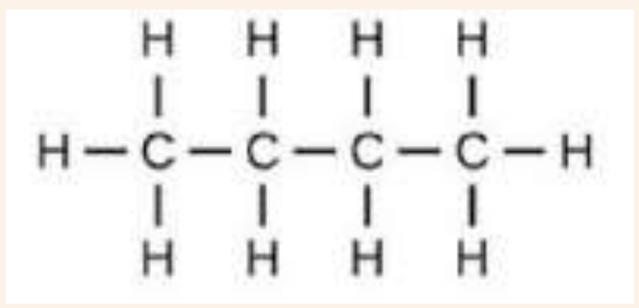
CBSE SQP 2019-2025

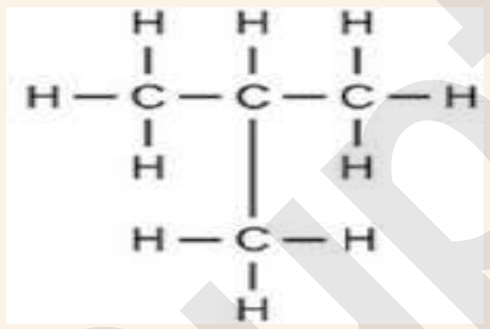
Ch.- Carbon and Its Compound

Section – A

Q.1. Define catenation. **(CBSE SQP 2019-20)**

Q.2. **Assertion:** Following are the structural isomers of butane. **(CBSE SQP 2019-20)**





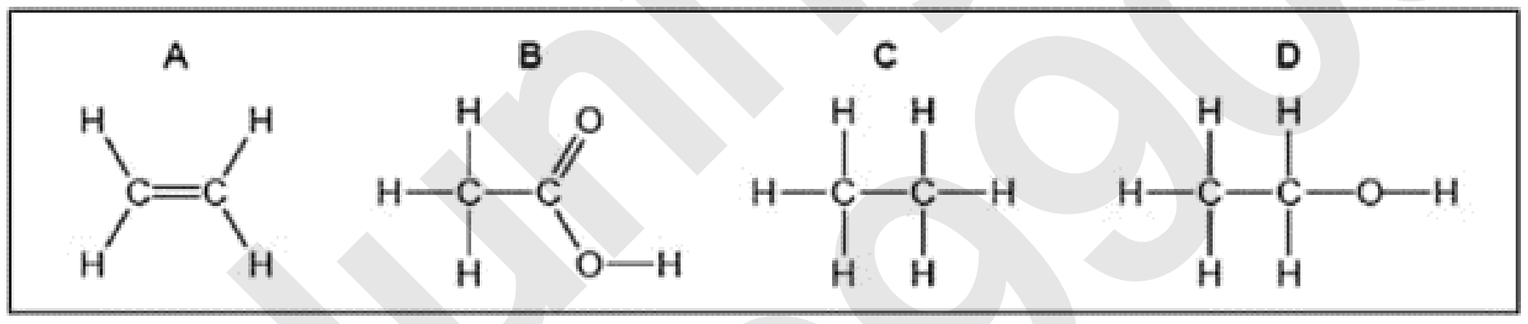
Reason: Structural isomers have the same molecular formula but they differ in their structures.

Q.3. Which of the following is not observed in a homologous series? Give reason for your choice.

a) Change in chemical properties b) Difference in -CH2 and 14u molecular mass

c) Gradation in physical propertiesd) Same functional group **(CBSE SQP 2020-21)**

Q.4. The formulae of four organic compounds are shown below. Choose the correct option:



(a) A and B are unsaturated hydrocarbons **(CBSE SQP 2022-23)**

(b) C and D are saturated hydrocarbons

(c) Addition of hydrogen in presence of catalyst changes A to C

(d) Addition of potassium permanganate changes B to D

Q.5. **Assertion (A):** C_4H_8, C_4H_6 and C_4H_{10} are members of the same homologous series

Reason (R): $C4H_8$, $C4H_6$, C_3H_4, C_3H_6, C_2H_4, C_2H_2 are unsaturated hydrocarbons.

(CBSE SQP 2025-26)

Section – B

Q.6. Give a test that can be used to confirm the presence of carbon in a compound. With a valency of 4, how is carbon able to attain noble gas configuration in its compounds?

OR

The number of carbon compounds is more than those formed by all other elements put together. Justify the statement by giving two reasons. **(CBSE SQP 2020-21)**

Section – D

Q.7. The formulae of four organic compounds are given below: **(CBSE SQP 2019-20)**

A	B	C	D
C_2H_4	CH_3COOH	C_2H_5OH	C_2H_6

(i) Which one of these compounds A, B, C or D is a saturated hydrocarbon?

(ii) Identify the organic acid and give its structural formula.

(iii) Which of the above compounds when heated at 443K in the presence of concentrated H_2SO_4 forms ethene as the major product? What is the role played by concentrated H_2SO_4 in this reaction? Also write the chemical equation involved.

(iv) Give a chemical equation when B and C react with each other in presence of concentrated H_2SO_4. Name the major product formed and mention one of its important use.

Q.8. Shristi heated Ethanol with a compound A in presence of a few drops of concentrated sulphuric acid and observed a sweet smelling compound B is formed. When B is treated with sodium hydroxide it gives back Ethanol and a compound C. **(CBSE SQP 2022-23)**

(a) Identify A and C

(b) Give one use each of compounds A and B.

(c) Write the chemical reactions involved and name the reactions.

OR

(a) What is the role of concentrated Sulphuric acid when it is heated with Ethanol at 443 K. Give the reaction involved.

(b) Reshu by mistake forgot to label the two test tubes containing Ethanol and Ethanoic acid. Suggest an experiment to identify the substances correctly? Illustrate the reactions with the help of chemical equations.

Q.9. Rehmat classified the reaction between Methane and Chlorine in presence of sunlight as a substitution reaction. Support Rehmat's view with suitable justification and illustrate the reaction with the help of a balanced chemical equation. **(CBSE SQP 2023-24)**

OR

Raina while doing certain reactions observed that heating of substance 'X' with vinegar like smell with a substance 'Y' (which is used as an industrial solvent) in presence of conc. Sulphuric acid on a water bath gives a sweet-smelling liquid 'Z' having molecular formula $C_4H_8O_2$. When heated with caustic soda (NaOH), 'Z' gives back the sodium salt and the compound 'Y'.

Identify 'X', 'Y', and 'Z'. Illustrate the changes with the help of suitable chemical equations.

Attempt either option A or B

Q.10. (a)

(i) "Keerthi thinks that Substitution reaction occurs in saturated Hydrocarbons, on the contrary Krishi thinks, it occurs in unsaturated Hydrocarbons." Justify with valid reasoning whose thinking is correct. **(CBSE SQP 2024-25)**

(ii) "Methane and Propane and their Isomers are used as fuels" Comment. Draw the electron dot structure of the immediate lower homologue of Propane. Give any two characteristics of homologues of a given homologous series.

(iii) A mixture of oxygen and ethyne is burnt for welding. Can you predict why a mixture of ethyne and air is not used?

OR

(b)

(i) 'A' & 'B' are sodium salts of long-chain carboxylic acid and long chain Sulphonic acid respectively. Which one of A or B will you prefer as a cleansing agent while using underground water (hand pump water)? Give the reason for your answer.

(ii) Elaborate on the process of cleansing action. Illustrate micelle with the help of labelled diagram.

(iii) Write the chemical equation of the preparation of soap from an ester CH_3COOCH_3. What is the name of this process?

Q.11. **Attempt either option A or B.**

A. A hydrocarbon with the formula CxHy undergoes complete combustion as shown in the following equation: **(CBSE SQP 2025-26)**

$2C_xH_y + 9O_2 \rightarrow 6CO_2 + 6H_2O$.

(a) What are the values of 'x' and 'y'?

(b) Give the chemical (IUPAC) name of the hydrocarbon.

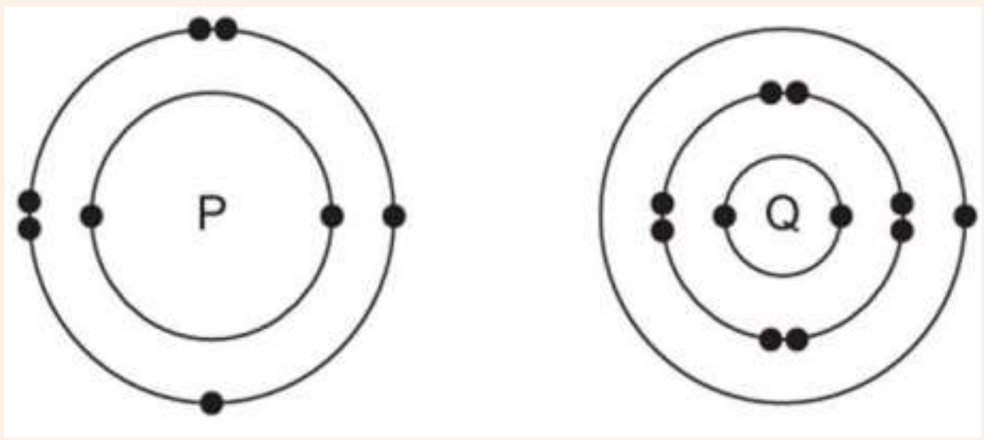
(c) Draw its electron dot structure.

(d) Name the alcohol which on heating with conc. H_2SO_4 will produce the above hydrocarbon C_xH_y.

(e) Write a balanced chemical equation for the reaction of C_xH_y with hydrogen gas in presence of Nickel.

OR

B. The electronic structures of atoms P and Q are shown below



Based on the information given above, answer the following questions:

(a) If P and Q combine to form a compound, what type of bond is formed between them?

(b) Give the chemical formula of the compound formed.

(c) The compound so formed is dissolved in water. Is the resultant solution acidic or basic in nature? Justify your answer.

(d) Write the chemical equation for the reaction between 'Q' and ethanol.

(e) What will be the formula of the compound formed when 'P' undergoes bonding with carbon?

Section – E

Q.12. The table given below shows the hints given by the quiz master in a quiz.

S. No	Hint
(i)	Substance ' C' is used as a preservative.
(ii)	C' has two carbon atoms; C' is obtained by the reaction of 'A' in presence of alkaline Potassium permanganate followed by acidification.
(iii)	Misuse of A' in industries is prevented by adding Methanol, Benzene, and pyridine to A' 'F' is formed on heating 'A' in presence of conc Sulphuric acid.
(iv)	'F' reacts with Hydrogen gas in presence of Nickel and Palladium catalyst.

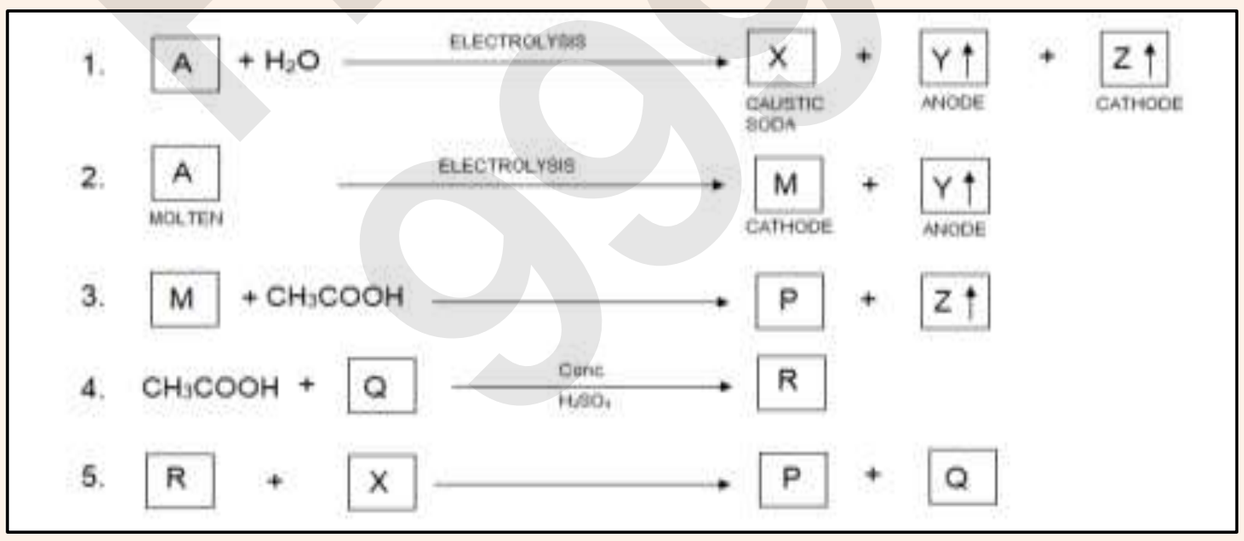
Based on the above hints answer the following questions **(CBSE SQP 2023-24)**

(a) Give the IUPAC names of A and F

(b) Illustrate with the help of chemical equations the changes taking place. (A→C and A→F). **OR**

Name the chemical reactions which occur in steps 2 and 5. Identify the compounds formed in these steps if ' A' is replaced with its next homologue.

Q.13.



(a) Derive the names of A, Y, Z, M, P & R **(CBSE SQP 2024-25)**

Attempt either subpart B or C.

(b) Improvise an activity to test Z.

OR

(c) Name the process in which compounds X, Y & Z are formed from A. Justify your response.